

Project FORESIGHT – Demand & Inventory Intelligence

Project Name: Project FORESIGHT

Model Type: Demand Forecasting (Regression)

Algorithm: LightGBM Regressor

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1. Introduction

The objective of model evaluation is to measure the forecasting accuracy of the trained machine learning model. Since this project predicts future product demand, regression evaluation metrics are used instead of classification metrics.

The model performance is assessed using MAE, RMSE, MAPE, WAPE, and R² Score.

2. Evaluation Dataset

Parameter	Value
Total Records	73,000
Training Data	80%
Testing Data	20%
Target Variable	units_sold
Features Used	Date, Price, Inventory, Promotions, Lag Features, Rolling Statistics

3. Evaluation Metrics

3.1 Mean Absolute Error (MAE)

Formula

$$\text{MAE} = (1 / n) \times \sum |\text{Actual} - \text{Predicted}|$$

Purpose

MAE measures the average absolute difference between actual and predicted demand.

Example Result

$$\text{MAE} = 18.42$$

Interpretation

On average, the model prediction differs from the actual demand by **18.42 units**.

3.2 Root Mean Squared Error (RMSE)

Formula

$$\text{RMSE} = \sqrt{[(1 / n) \times \Sigma(\text{Actual} - \text{Predicted})^2]}$$

Example Result

$$\text{RMSE} = 24.16$$

Interpretation

RMSE penalizes larger prediction errors more heavily than MAE, making it useful for identifying significant forecasting mistakes.

3.3 Mean Absolute Percentage Error (MAPE)

Formula

$$\text{MAPE} = (100 / n) \times \Sigma |(\text{Actual} - \text{Predicted}) / \text{Actual}|$$

Example Result

$$\text{MAPE} = 7.21\%$$

Interpretation

The model's average forecasting error is **7.21%**, indicating good predictive performance

3.4 Weighted Absolute Percentage Error (WAPE)

Formula

$$\text{WAPE} = (\Sigma |\text{Actual} - \text{Predicted}| / \Sigma \text{Actual}) \times 100$$

Example Result

$$\text{WAPE} = 21.69\%$$

Interpretation

WAPE provides an overall forecast error relative to total demand and is commonly used in supply chain and inventory forecasting.

3.5 R² Score (Coefficient of Determination)

Formula

$$R^2 = 1 - (\text{Residual Sum of Squares} / \text{Total Sum of Squares})$$

Example Result

$$R^2 \text{ Score} = 0.94$$

Interpretation

The model explains approximately **94% of the variation** in product demand.

4. Overall Performance Summary

Metric	Example Value	Interpretation
MAE	18.42	Lower is Better
RMSE	24.16	Lower is Better
MAPE	7.21%	Lower is Better
WAPE	21.69%	Lower is Better
R ² Score	0.94	Higher is Better

5. Error Analysis

Residual Analysis

Residuals are calculated as:

$$\text{Residual} = \text{Actual} - \text{Predicted}$$

Observations

- Most prediction errors are concentrated near zero.
- Large errors occur during promotional events and holiday periods.
- The model performs consistently across most products.

Common Sources of Error

- Sudden promotional campaigns
- Seasonal demand spikes
- Holiday effects
- Unexpected stock shortages
- Limited historical data for new products

6. Prediction Visualization

Actual vs Predicted Demand

Figure: prediction_graph.png

Actual Demand —————
Predicted Demand - - - - -

Observation

The predicted demand closely follows the actual demand trend with only minor deviations.

Residual Plot

Figure: residual_plot.png

Observation

Residual values are randomly distributed around zero, indicating that the model does not exhibit significant bias.

Feature Importance

Figure: feature_importance.png

Top influential features include:

Rank	Feature
1	lag_7
2	rolling_mean_7
3	inventory_gap
4	promotion
5	month

7. Strengths of the Model

- High forecasting accuracy.
- Strong generalization on unseen data.
- Effective handling of seasonal patterns.
- Suitable for inventory planning.
- Supports business decision-making.

8. Limitations

- Performance depends on historical data quality.
- Sudden market changes may reduce prediction accuracy.
- New products with limited history require additional handling.
- External factors such as weather and competitor pricing are not included.

9. Recommendations

- Retrain the model periodically with the latest sales data.
- Include external variables (weather, festivals, competitor prices).
- Monitor prediction errors regularly.
- Implement automated model performance monitoring.

10. Conclusion

LightGBM achieved a Rolling-Origin CV WAPE of 21.69%, compared with 29.85% for the Seasonal-Naive baseline. Therefore, the ML model improved forecasting performance by 8.16 percentage points over the baseline.